ELECTENG 310 Team 9

Water level detection device PoC

SPECIFICATION

ELECTRICAL PARAMETERS (3 marks)

Input current I (A)	0.153A
Input voltage V (V)	12V
Input power P (W) (2 marks)	1.836W
Frequency range $f_1 \rightarrow f_2$ (Hz) (1 mark)	3kHz - 25kHz
Wavelength (nm)	850nm

RANGE (3 marks)

Transmission range (m): 2m (3 marks) 1m (2 marks) 0.2m (1 mark)	Our range is 2.3m

STABILITY (4 marks)

Stable signal (no fluctuations) (2 marks)	The 1V is quite noisy but high water level is stable
Resilience to ambient light (1 mark)	Shining a light into the cone piece will affect the receiver Light above or to the side does not affect receiver
Resilience to unstable V_{cc} (± 10%) (1 mark)	Low tank gives 0.93V at Vcc=10.8V Low tank gives 0.93V at Vcc=13.2V High tank is stable

ACCURACY OUTPUT VOLTAGE (4 marks)

"No water" signal 1V± 20% (2 marks) 1V± 40% (1 mark)	0.95V - 0.96V
"Full tank" signal 3V± 10% (2 marks) 3V± 20% (1 mark)	2.76V

ACCURACY 2 (3 marks)

Accuracy at 10 % water 1.856V 1.876V	ΔC = 3 (pF) ΔV = 0.02(V) Accuracy 98.92 (%)

FAULT MANAGEMENT (1 mark)

Loss of transmission signal (V)	3V when no signal, this is outside of our range of 0.95V - 2.76V

ACCESSORIES (2 marks)

Signal capture device (Yes / No)	Yes we have light reflecting cones (reflector) to help concentrate light into the photodiode